

Assignment 1

We have a set of 1,000,000 documents (text format) and we want to:

- **Extract structured metadata/entities** for authors and references as well as key methods and/or concepts (either free text or linked to a given taxonomy or ontology)
 - **Identify relationships** among the above-mentioned entities (e.g., similar publications, similar authors) and **address the following information retrieval tasks**:
 - Pub2Pub: Given a publication to suggest other related publications
 - Pub2Author: Given a publication to suggest related Authors
 - Concept or Keyword based search: Given a set of keywords or concepts to suggest related articles
1. **Methods & data pipelines**: Briefly describe the steps and the methods you would follow to address the above-mentioned problems (e.g., the type of ML tasks at hand, pre-processing pipelines, evaluation processes, whether you need manual annotated data for some tasks and of what kind, potential tools and/or frameworks you would use, potential risks & challenges)
 2. **Scalability**: Assuming that you need to scale such methods & pipelines on 100 million publications, describe the additional challenges you would face and how you would try to address them.

Assignment 2

Given the following dataset of documents with their associated labels

(<https://drive.google.com/file/d/1H0Hmkcliv4FYnqHxwCP7C7gmxGzk5wy0/view?usp=sharing>), you need to build a classifier (in Java or Python) which classifies each instance into all possible labels it belongs to (preferably using a DL model).

The dataset is stored in a JSON file and each instance is a dictionary with two keys: *sentence* which contains the text of each instance in a string and *labels* which contains a list with all the labels of each instance. You can use any library / method you want.

Hint: The text has been already preprocessed (stemming, stopwords removal, lower case, etc..).

You need to provide a short description of your solution as well as a report along with your code.